Entropic Ledger Time: An SMFT Interpretation of Cold-Atom Tests of the Problem of Time

How a bright-sector "mini-universe" turns entropy exchange into internal time through boundary, gate, trace, residual, and ledger

---

Front Note — Interpretation, Not Replacement

This article interprets the academic paper **"Testing the problem of time with cold atoms"** through the lens of **Semantic Meme Field Theory** (SMFT), especially the later frameworks of **declared disclosure**, **imaginary time as admissibility depth**, and the **residual-to-ledger cycle**.

The claim is **not** that the cold-atom experiment proves SMFT.

The claim is narrower and more useful:

The experiment provides a striking physical analogue of SMFT ledger-time: a subsystem acquires internal time when boundary-mediated exchange becomes measurable trace and is ordered inside a declared observation protocol.

The academic paper remains a physics result. SMFT supplies an interpretive grammar.

In compact form:

```
(0.1) Closed field → declared subsystem → boundary gate → entropy-bearing trace → ledger → internal time.
```

Or even shorter:

```
(0.2) Time begins when a world can admit change into trace.
```

Abstract

The academic paper **"Testing the problem of time with cold atoms"** realizes a controlled cold-atom analogue of relational time. A Bose–Einstein condensate is prepared as a well-isolated many-body system in a conservative trap, then partitioned by an optical barrier into an observed **bright sector** and an unobserved **dark sector**. Although the total system is treated as effectively closed and governed by a time-independent Hamiltonian, the bright sector can be ordered by an internally constructed **entropic time** derived from measurable entropy exchange. The paper shows that this entropic time robustly orders the bright-sector dynamics and supports an effective Schrödinger equation parameterized by the internal time variable.

This article reads that result through SMFT. In SMFT, time is not assumed as a primitive background container. Time emerges when a field is declared, projected, gated, traced, residualized, and ordered into a ledger. The declared-disclosure framework expresses this as:

```
(0.3) Σ₀ → Declare_P → Σ_P → Ô_P → Gate_P → Trace_P + Residual_P → Ledger_P → Time_P.
```

The cold-atom experiment maps naturally onto this grammar. The total condensate functions as the closed field. The bright sector is the declared observed world. The dark sector is residual relative to the bright-sector ledger. The optical barrier is the boundary gate. Entropy exchange is trace admission. Entropic time is the order of entropy-bearing traces.

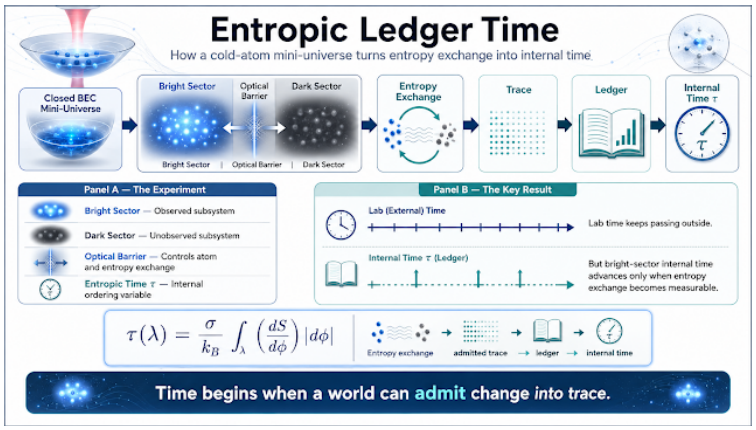
The main thesis of this article is therefore:

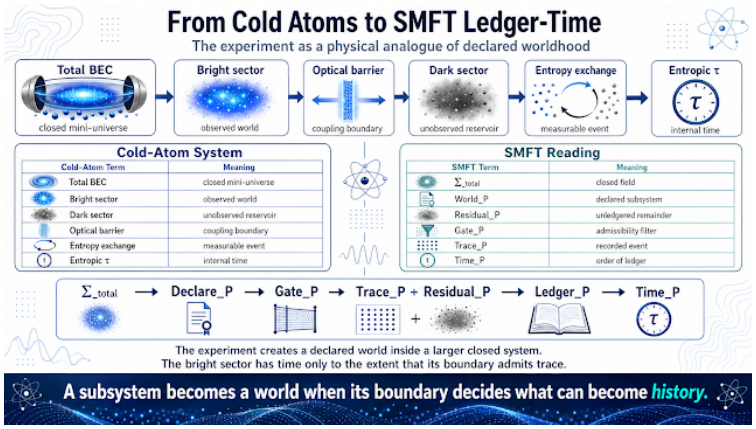
```
(0.4) Entropic time is ledger time generated by boundary-mediated entropy trace.
```

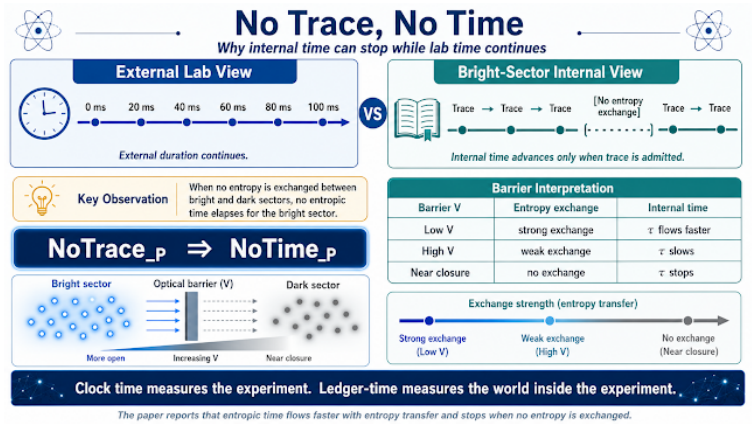
This interpretation clarifies why an internal clock cannot be merely a changing variable. A useful internal clock must be trace-bearing, monotonic enough, boundary-relevant, and reproducible under a declared protocol. The cold-atom paper shows precisely this: the scalar-like clock variable ϕ alone is not globally sufficient in a recollapsing system, while entropy-weighted ordering provides a robust internal arrow.

The result is philosophically important because it supports a disciplined version of a broader claim:

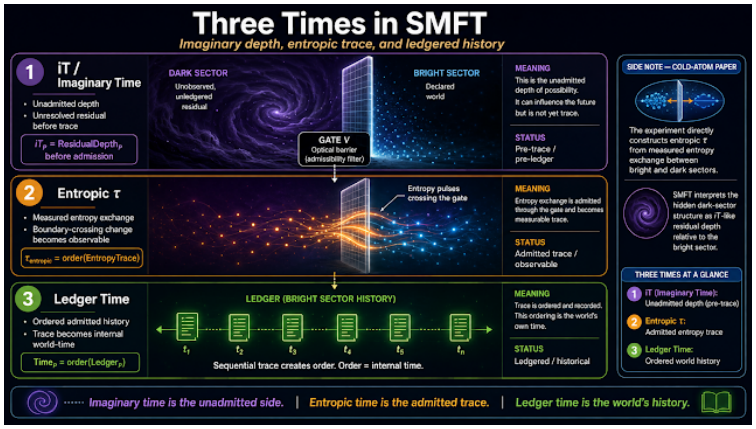
```
(0.5) Internal time is not external duration imported into a subsystem; internal time is the ordered disclosure of what that subsystem can record as its own history.
```











---

0. Reader's Guide: What This Article Is and Is Not

0.1 What this article is

This article is a conceptual bridge between three things:

```
(0.6) Cold-atom relational time experiment.
(0.7) SMFT declared-disclosure theory of time.
(0.8) Residual-to-ledger ontology of world formation.
```

The cold-atom paper supplies a real experimental system. SMFT supplies a grammar of boundary, projection, gate, trace, residual, and ledger. The residual-to-ledger framework supplies a deeper ontology of how a world begins when previously unledgered material becomes admissible trace.

The article's core interpretation is:

```
(0.9) A subsystem has internal time only to the extent that it can admit, record, and order trace-bearing events.
```

0.2 What this article is not

This article does not claim:

```
(0.10) The cold-atom experiment proves SMFT.
(0.11) Entropic time is identical to all forms of physical time.
(0.12) The dark sector is literally metaphysical residual.
(0.13) The laboratory mini-universe is literally a cosmological universe.
```

The correct level of claim is structural:

```
(0.14) The experiment realizes a physical analogue of ledger-time.
```

That is already significant.

0.3 Why this matters

The problem of time often appears as a question about clocks:

```
(0.15) Which variable can act as time?
```

SMFT reframes the question:

```
(0.16) Under what declaration can change become trace, and under what ledger can trace become time?
```

This shift is important. A changing variable is not automatically a clock. A clock must support ordered history.

---

1. The Academic Finding: A Cold-Atom Testbed for the Problem of Time

1.1 The experiment in plain language

The cold-atom paper realizes a Bose–Einstein condensate in a conservative optical trap. A thin optical barrier partitions the condensate into two regions:

```
(1.1) Bright sector = observed sector.
(1.2) Dark sector = unobserved sector.
```

The barrier allows atoms to cross depending on its height. When atoms begin to populate the bright sector, the paper calls this analogue event a "big bang." When atoms return to the dark sector, it calls the event a "big crunch." Figure 1 of the paper shows absorption images of the condensate evolving in external lab time, with the barrier separating the unobserved dark sector from the observed bright sector.

In ordinary laboratory language, the system simply oscillates in time.

But the paper asks a deeper question:

```
(1.3) Can the observed sector be ordered without using external lab time?
```

This is why the experiment matters. It does not merely observe motion. It tests whether an **internal time** can be constructed from within a subsystem.

1.2 The Wheeler–DeWitt motivation

The paper is motivated by the problem of time in Wheeler–DeWitt-type quantum gravity. In such settings, the total state is not naturally sequenced by an external time parameter. If the universe as a whole has no outside clock, then time must be recovered relationally, from correlations between internal degrees of freedom.

The cold-atom experiment does not solve quantum gravity. It builds an analogue system where similar questions can be operationally tested:

```
(1.4) Total isolated system → no need for external time inside the description.
(1.5) Observed subsystem → possible internal ordering through relational variables.
```

The paper explicitly describes its system as a well-isolated cold-atom platform with a time-independent Hamiltonian, used to probe relational time constructions. It then partitions the system into dark and bright sectors and constructs an entropic time for the bright sector.

1.3 The bright-sector Hamiltonian

The paper writes the total Hamiltonian schematically as:

```
(1.6) Ĥ = Ĥ_bright + Ĥ_dark + Ĥ_coupling.
```

This is already a boundary ontology.

The total system is not simply "one thing." It is treated as:

```
(1.7) total system = observed sector + unobserved sector + coupling gate.
```

From the SMFT viewpoint, this is the beginning of a declared world. A world appears only after a boundary separates what is visible from what remains residual.

1.4 Entropic time

The paper defines entropic time for the bright sector as:

```
(1.8) τ(λ) = σ/k_B ∫_λ (dS/dφ)|dφ|.
```

Here:

```
(1.9) S = entropy in the bright sector.
(1.10) φ = internal scalar-like clock field, experimentally represented by the bright-sector center-of-mass coordinate.
(1.11) σ = arbitrary entropic time unit.
(1.12) λ = trajectory of φ within the bright sector.
```

The definition is designed so that the arrow of time remains meaningful when entropy and the clock-field trajectory move together. The paper emphasizes that this entropic time is built from thermodynamic entropy rather than from a Hamilton–Jacobi function, and that the experiment directly tests whether it meaningfully orders

the data.

The key empirical finding is that τ grows monotonically almost everywhere, and that its slope is set by entropy flow in the bright sector. Entropic time flows faster when entropy is transferred in or out, and stops when no entropy is exchanged with the dark sector. The paper also reports that the total entropy of the mini-universe remains constant within error bars.

This is the experimental seed of the SMFT interpretation.

---

2. The SMFT Reframing: Time Is Not Change, but Ordered Trace

2.1 The ordinary reading

The ordinary reading of the cold-atom experiment is:

```
(2.1) Entropy exchange provides an internal arrow of time.
```

This is correct, but incomplete.

It says what the clock is made from, but not why it counts as time.

SMFT adds a deeper interpretation:

```
(2.2) Entropy exchange counts as internal time only after it becomes admitted trace in a declared subsystem.
```

This distinction matters. There can be change without internal time. There can be motion without world-history. There can be hidden dynamics without ledgered consequence.

2.2 The SMFT time chain

The declared-disclosure framework states that a pre-time field must be declared before it can be filtered, projected, gated, traced, residualized, and stabilized as time. It introduces the declared protocol:

```
(2.3) P = (B, Δ, h, u).
```

where:

```
(2.4) B = boundary.
(2.5) Δ = observation or aggregation rule.
(2.6) h = time or state window.
(2.7) u = admissible intervention family.
```

It then defines the declared field:

```
(2.8) Σ_P = Declare(Σ₀ | q, ϕ, P).
```

and the gauged disclosure operator:

```
(2.9) 𝔇_P = UpdateTrace_P ∘ Gate_P ∘ Ô_P ∘ Declare_P.
```

The resulting definition of time is:

```
(2.10) Time_P = order(𝔇_P(Σ₀)).
```

The declared-disclosure article explicitly states that time is not merely ledgered filtration, but ledgered disclosure of a declared field.

2.3 The cold-atom translation

The cold-atom paper maps cleanly onto this chain:

SMFT term	Cold-atom realization
Σ0	total closed condensate system
P	experimental declaration protocol
B	optical barrier separating bright and dark sectors
Δ	absorption imaging and density-profile measurement
h	120 ms experimental sequence with 2 ms sampling
φ	measured features: N, ϕ, Σ, S
Ô_P	projection onto bright-sector observables
Gate_P	barrier-mediated atom and entropy exchange
Trace_P	recorded entropy-bearing bright-sector events
Residual_P	dark-sector remainder and unobserved correlations
Ledger_P	ordered sequence of bright-sector records
Time_P	entropic ordering τ

This mapping gives the article its central equation:

```
(2.11) τ_bright = order(EntropyTrace_bright).
```

Or more generally:

```
(2.12) Time_P = order(AdmittedTrace_P).
```

2.4 Why change alone is not enough

The scalar-like variable ϕ changes. But in a recollapsing system it is not globally monotonic. The paper explicitly notes that ϕ cannot act as a global time coordinate because it becomes ambiguous in a recollapsing bright sector.

SMFT interprets this as a general clock-quality principle:

```
(2.13) A changing variable is not yet time.
```

A good internal time variable must satisfy stronger conditions:

```
(2.14) ClockQuality = Monotonicity + Traceability + BoundaryRelevance + Reproducibility.
```

The entropic time τ is stronger than ϕ alone because it is tied to trace-bearing entropy exchange across the declared boundary.

In plain language:

```
(2.15) φ tells us where the bright sector is.
(2.16) τ tells us what the bright sector has admitted into history.
```

---

3. The Bright Sector as a Declared World

3.1 A subsystem becomes a world only under declaration

The bright sector is not automatically a world. It becomes world-like because the experiment declares:

```
(3.1) This side of the barrier is the observed sector.
(3.2) These variables count as observables.
(3.3) This entropy definition counts as internal trace.
(3.4) This ordering counts as internal time.
```

This is exactly why SMFT begins with declaration. A world is not merely a region of matter. A world is a region under a protocol of observability, admissibility, trace, and ledger.

The declared-disclosure framework states that a viewpoint must become a declaration: it must specify what counts as inside, what counts as observable, which horizon is being used, what interventions are admissible, what baseline is assumed, what feature map detects structure, what gate commits projection into trace, and what residual remains after closure.

The cold-atom experiment is a concrete physical example of this principle.

3.2 The optical barrier as boundary and gate

The optical barrier has two roles.

First, it is a boundary:

```
(3.5) B_V separates bright from dark.
```

Second, it is a gate:

```
(3.6) Gate_V determines how much atom and entropy exchange becomes admissible to the bright sector.
```

The barrier height V is therefore not merely a potential parameter. In the SMFT interpretation, it is a **gate-strength parameter**.

Lower V means easier passage:

```
(3.7) Low V → more exchange → more entropy trace → faster entropic time.
```

Higher V means harder passage:

```
(3.8) High V → less exchange → less entropy trace → slower entropic time.
```

The paper reports precisely this: higher barrier values progressively reduce entropy exchange, causing entropic time to flow more slowly; near $V \simeq 1$, the bright sector approaches a heat-death-like state where entropic time stops.

3.3 The dark sector as residual

In the experiment, the dark sector is unobserved relative to the bright-sector analysis. It is not absent. It remains part of the total system.

SMFT calls this kind of remainder **residual**:

```
(3.9) Residual_P = what remains unresolved, unadmitted, unobserved, or unledgered under protocol P.
```

The declared-disclosure framework emphasizes that disclosure produces both trace and residual:

```
(3.10) Disclosure_P = Trace_P + Residual_P.
```

Trace is what becomes committed; residual is what remains unresolved. The framework states that trace stabilizes history while residual preserves unfinished possibility.

The cold-atom system gives this abstract principle a physical form:

```
(3.11) Bright trace = entropy-bearing events admitted into measurement.
(3.12) Dark residual = unobserved sector and hidden relational structure.
```

3.4 Residual is not nothing

The Residual-to-Ledger framework strongly warns against treating residual as nothing. It defines residual as unadmitted, unresolved, unledgered, or unconverted remainder, and distinguishes ordinary residual from generative residual that remains relation-rich, detachable, filterable, and trace-capable.

This matters for the cold-atom paper because the dark sector is not passive emptiness. It controls the bright sector's internal time through exchange.

In SMFT terms:

```
(3.13) DarkResidual → Gate_V → EntropyTrace → BrightLedger → BrightTime.
```

The bright sector does not generate time alone. Its internal time emerges from its relation to what it does not fully contain.

This is one of the deepest lessons of the experiment.

---

4. Entropic Time as Ledger Time

4.1 The academic formula

The paper's entropic time formula is:

```
(4.1) τ(λ) = σ/k_B ∫_λ (dS/dφ)|dφ|.
```

This formula is more subtle than a simple entropy clock.

It does not merely say:

```
(4.2) τ = S.
```

It says that internal time is accumulated along a trajectory of the clock-field variable ϕ, weighted by entropy change. The absolute trajectory term $|d\phi|$ helps preserve an oriented progression when entropy and ϕ cooperate in the chosen construction.

This is why the paper can use τ to order expansion and recollapse even when ϕ alone is not globally reliable.

4.2 The SMFT formula

SMFT rewrites the idea as:

```
(4.3) Trace_k = Gate_P(EntropyExchange_k).
```

Then:

```
(4.4) L_{k+1} = Update(L_k, Trace_k, Residual_k).
```

And:

```
(4.5) Time_P = order(L_P).
```

For the bright sector:

```
(4.6) τ_bright = order({Gate_V(ΔS_k)}).
```

In words:

> **Entropic time is not entropy itself. Entropic time is the ordered ledger of entropy-bearing events admitted through the boundary gate.**

This is the article's key definition:

```
(4.7) EntropicLedgerTime_P = order(EntropyTrace_P).
```

4.3 Why no entropy exchange means no internal time

The paper reports a crucial observation: for low barrier values, the bright sector cycles from analogue big bang to analogue big crunch, but no entropic time elapses between a big crunch and the subsequent big bang because no entropy is exchanged there.

This is the strongest experimental analogue of the SMFT thesis.

In lab time:

```
(4.8) external duration continues.
```

In bright-sector entropic time:

```
(4.9) no entropy exchange → no entropic time increment.
```

In SMFT language:

```
(4.10) NoTrace_P ⇒ NoTime_P.
```

This does not mean nothing exists. It means nothing becomes admitted into the bright-sector ledger.

The distinction is decisive:

```
(4.11) Existence is not the same as ledgered time.
(4.12) Hidden dynamics are not the same as admitted history.
(4.13) Lab duration is not the same as subsystem time.
```

4.4 Time is admitted history

The Residual-to-Ledger framework says:

```
(4.14) A world is not everything that happens.
(4.15) A world is what a boundary allows to become history.
```

It further states that the gate decides what becomes trace, trace enters a ledger, and the ledger creates time because ordered trace constrains the future.

The cold-atom paper gives a clean laboratory analogue:

```
(4.16) The bright sector is not everything that happens in the condensate.
(4.17) The bright sector is what the bright-sector boundary allows to become measured history.
```

Therefore:

```
(4.18) BrightTime = order(BrightHistory).
```

And bright history is not all motion. It is admitted entropy trace.

Interim Summary

So far, the argument is:

```
(4.19) The cold-atom experiment creates a closed total system.
(4.20) It declares a bright observed sector and a dark residual sector.
(4.21) The optical barrier functions as boundary and gate.
(4.22) Entropy exchange becomes trace when admitted into the bright-sector record.
(4.23) The ordered sequence of such traces becomes entropic time.
(4.24) When no entropy trace is admitted, no internal entropic time elapses.
```

This is why the experiment is so important for SMFT:

```
(4.25) It realizes, in laboratory form, the claim that time is ordered admitted trace.
```

Continue from Section 5: The Dark Sector as Residual and the Residual-to-Ledger Cycle.

5. The Dark Sector as Residual and the Residual-to-Ledger Cycle

5.1 The dark sector is not absence

In the cold-atom experiment, the **dark sector** is the unobserved side of the partition. It is not physically absent. It is part of the same closed many-body system. The paper explicitly treats the total system as a bright sector, a dark sector, and a coupling term:

(5.1) $\hat{H} = \hat{H}_{bright} + \hat{H}_{dark} + \hat{H}_{coupling}$.

The bright sector is then studied as the observed subsystem, while the dark sector remains unobserved relative to that description. The barrier determines how much matter and entropy can cross between the two sectors.

This is exactly where the **Residual-to-Ledger** interpretation becomes useful.

In ordinary language, we might say:

(5.2) The bright sector is observed.
(5.3) The dark sector is unobserved.

But SMFT sharpens this into:

(5.4) The bright sector is ledgered.
(5.5) The dark sector is residual relative to the bright-sector ledger.

This does not mean the dark sector is unreal. It means the dark sector is not currently part of the bright sector's admitted history.

The dark sector is therefore not "nothing." It is **unledgered remainder**.

5.2 Residual as hidden but world-relevant structure

The Residual-to-Ledger framework defines residual as what remains unadmitted, unresolved, unledgered, or unconverted under a given boundary and protocol. It also emphasizes that residual is not always waste. Sometimes residual is danger. Sometimes it is noise. Sometimes it is future possibility. In its strongest form, residual can become seed for a new ledgered world if it remains relation-rich, detachable, filterable, and trace-capable.

In the cold-atom experiment, the dark sector functions as residual in a very concrete way:

(5.6) DarkResidual = unobserved but physically active remainder.

It remains outside the bright-sector ledger, but it still affects the bright sector through coupling.

Thus:

(5.7) Residual does not mean irrelevant.

The bright sector's internal time depends on exchange with the dark sector. When atoms and entropy cross the barrier, residual becomes trace. When no exchange occurs, residual remains outside the bright-sector ledger.

This gives the central cold-atom residual chain:

(5.8) DarkResidual → Gate_V → EntropyTrace → BrightLedger → BrightTime.

In words:

> **The dark sector is the hidden reservoir whose gate-mediated exchange allows the bright sector to have internal time.**

5.3 The barrier as residual filter

The barrier is more than a wall. It is a **filter of world-entry**.

In physical terms, the barrier height changes the coupling between sectors. The paper varies the height of the potential barrier and shows that this changes the entropy exchange between the sectors; the slope of entropic time depends on entropy flow, and entropic time stops when no entropy is exchanged.

In SMFT terms:

(5.9) Gate_V = admissibility filter between residual and trace.

Lower barrier:

(5.10) V low → residual crosses easily → trace accumulates → τ flows faster.

Higher barrier:

(5.11) V high → residual is blocked → trace slows → τ slows or stops.

At very high barrier, the bright sector stops receiving enough entropy-bearing exchange to maintain cyclic internal time. The paper describes the $V \simeq 1$ condition as a heat-death-like regime in which entropic time completely stops.

The SMFT version is:

(5.12) Gate closure → trace starvation → ledger stagnation → internal time stops.

This is a powerful physical analogue of ledger ontology.

5.4 Residual-to-ledger cycle in the bright sector

The Residual-to-Ledger framework gives a general cycle:

(5.13) Boundary → Gate → Trace → Ledger → Residual → Revision / Redeclaration.

It also states that time appears only after trace enters a ledger:

(5.14) Time_P = order(Ledger_P).

And, in its more developed form:

(5.15) PreTimeField_P → FilterDepth_P → Trace_P → Ledger_P → Time_P.

The framework explicitly identifies iTime not as a hidden clock, but as admissibility depth before trace, while real time is ordered ledgered consequence.

The cold-atom system instantiates a small physical cycle:

```
(5.16) Dark sector residual
→ barrier gate
→ bright-sector atom / entropy admission
→ measured trace
→ entropic ledger
→ internal time
→ recollapse
→ return to residual.
```

This is not merely a metaphor. It is a structural reading of the experiment's cycle.

The bright-sector "big bang" is not an absolute beginning. It is a **trace-entry event**.

The bright-sector "big crunch" is not absolute annihilation. It is a **ledger-episode closure**.

The next cycle begins when residual again becomes admissible.

---

6. Big Bang, Big Crunch, and Ledger Episodes

6.1 The experiment's mini-cosmological language

The paper uses "big bang" for the moment atoms begin populating the bright sector and "big crunch" for the moment the atoms return to the dark sector. Figure 1 marks these with blue and green stars. The paper is careful: this is an analogue system, not a literal universe. But the terminology is useful because the bright sector behaves like a small world-episode: it appears, expands, contracts, and closes.

SMFT interprets this as:

```
(6.1) BigBang_bright = first trace admission into bright ledger.
```

and:

```
(6.2) BigCrunch_bright = closure of bright ledger episode.
```

The "universe" of the bright sector begins not when the total condensate begins, but when atoms and entropy cross into the declared observed sector.

This is a key distinction.

The total system exists before the bright-sector big bang. But the bright-sector world does not yet have its own ledgered history.

6.2 Ledger episode structure

The bright-sector episode can be written as:

```
(6.3) LedgerEpisode_bright = FirstTrace → Expansion → Saturation → Recollapse → Closure.
```

Each term has a physical and SMFT meaning:

Stage	Cold-atom meaning	SMFT meaning
FirstTrace	atoms enter bright sector	world-entry event
Expansion	bright sector grows	ledgered world develops
Saturation	maximum extension	peak expressed trace
Recollapse	atoms return	trace-bearing structure withdraws
Closure	big crunch	episode ends / residual returns

The experiment's bright-sector episode therefore becomes a physical miniature of world-formation:

```
(6.4) A world begins when trace enters.
(6.5) A world develops while trace accumulates.
(6.6) A world closes when trace-bearing structure withdraws.
```

This is exactly the Residual-to-Ledger principle:

```
(6.7) A world is not everything that happens.
(6.8) A world is what a boundary allows to become history.
```

6.3 No singularity in the bright-sector ledger

The paper notes that the bright sector does not experience a true singularity in the big bang or big crunch because the center of the potential barrier is at $\phi = 0$ and ϕ is bounded by the barrier thickness.

This fits the SMFT interpretation beautifully.

The bright-sector big bang is not a metaphysical creation from nothing. It is a **gate-crossing event**.

The bright-sector big crunch is not absolute extinction. It is **return through the gate**.

Thus:

```
(6.9) BigBang_bright ≠ creation ex nihilo.
(6.10) BigBang_bright = admission of residual into ledger.
```

and:

```
(6.11) BigCrunch_bright ≠ absolute destruction.
(6.12) BigCrunch_bright = closure and residual return.
```

This is a safer and more precise ontology than saying a world begins from nothing.

The experiment gives us a vivid physical analogue:

```
(6.13) The beginning of a world may be a boundary event, not an absolute origin.
```

6.4 Lab time versus ledger episode time

In lab time, the whole oscillation proceeds continuously. But in entropic time, intervals with no entropy exchange do not count as internal bright-sector time.

This means the bright-sector episode has its own temporal grammar:

```
(6.14) External duration ≠ internal episode time.
```

or:

```
(6.15) Lab time measures the experiment.
(6.16) Entropic time measures the bright-sector world.
```

This distinction is central for the problem of time.

A world's internal time is not necessarily the same as the observer's external time. The paper uses lab time to validate the construction, but the construction itself aims to show that the bright sector can be ordered internally.

SMFT expresses this as:

```
(6.17) Time_P = order(Ledger_P), not order(ExternalClock).
```

---

7. The Effective Schrödinger Equation as Ledger-Time Dynamics

7.1 From timeless equation to entropic-time evolution

After constructing entropic time, the paper derives an effective Schrödinger equation parameterized by τ. Starting from a lab-time-independent equation, it obtains an entropic-time equation of the form:

```
(7.1) iℏ∂_τψ(τ,a) = Φ(τ)ψ(τ,a) + Λ(τ)H_geomψ(τ,a).
```

The paper explains that Φ is a global phase, while Λ acts as an entropy-dependent energy pump. Its derivative controls whether energy flows into or out of the scale degree of freedom a.

This is one of the most important technical bridges to SMFT.

In ordinary physics language:

```
(7.2) The bright-sector dynamics can be rewritten using entropic time.
```

In SMFT language:

```
(7.3) The declared world's dynamics can be rewritten in ledger-time.
```

7.2 Interpreting each term

SMFT translates the equation as follows:

Equation term	Physics role	SMFT interpretation
ψ(τ,a)	bright-sector wavefunction	state of declared bright world
a / Σ	scale-like degree of freedom	size of the ledgered world-expression
H_geom	geometric Hamiltonian term	internal geometry of the declared world
Φ(τ)	global phase	phase bookkeeping
Λ(τ)	entropy-dependent energy pump	residual-admission pump
∂τΛ > 0	energy flows into a	residual exchange energizes world-expansion
∂τΛ < 0	energy is sucked out	residual exchange withdraws world-expression

This yields:

```
(7.4) LedgerTimeDynamics = PhaseBookkeeping + ResidualModulatedGeometry.
```

The most important term is Λ(τ).

The paper says ordinary quantum mechanics applies locally when Λ varies slowly, while strictly unitary Schrödinger dynamics is recovered only when entropy flow is zero. It also notes that the equation is not well defined when there is no entropy, and therefore no time, flow.

SMFT interprets this as:

```
(7.5) Without trace flow, the ledger-time equation loses its time parameter.
```

This is exactly the ledger thesis again.

No trace flow, no internal time.

No internal time, no time-dependent bright-sector dynamics.

7.3 Λ as residual-admission pump

The paper describes Λ as an entropy-dependent energy pump. SMFT gives this a structural name:

```
(7.6) Λ(τ) = ResidualAdmissionPump_P.
```

This pump measures how the unobserved sector modulates the observed sector through entropy-bearing exchange.

So:

```
(7.7) ∂τΛ > 0 → residual admission feeds internal geometry.
(7.8) ∂τΛ < 0 → residual withdrawal drains internal geometry.
(7.9) ∂τΛ ≈ 0 → local ordinary dynamics.
```

The result is conceptually elegant:

> **The effective Hamiltonian of the bright world is not only an internal law. It is internal law under residual exchange.**

That is very close to SMFT's broader claim that world-dynamics are never merely intrinsic; they are shaped by boundary, gate, trace, and residual.

7.4 Local quantum mechanics as stable ledger regime

The paper notes that when Λ varies slowly, Eq. (6) reduces to a Schrödinger equation with a slowly varying effective Hamiltonian, so ordinary quantum mechanics applies locally with corrections controlled by $\partial\tau\Lambda$.

SMFT can read this as a general principle:

```
(7.10) Stable ledger flow → ordinary local law.
(7.11) Rapid residual exchange → time-dependent correction.
```

This suggests a broader ontology:

```
(7.12) Law is what internal dynamics look like when ledger conditions are stable.
```

When the boundary and residual exchange are steady, the declared world appears governed by local stable laws. When residual admission changes, the effective law changes with it.

This is not a rejection of physics. It is a meta-interpretation of when subsystem laws become clean.

---

8. Imaginary Time, Entropic Time, and Residual Depth

8.1 The experiment constructs τ, not iT

The cold-atom paper constructs an **entropic internal time τ**. It does not directly measure SMFT imaginary time iT.

This distinction is important.

In the SMFT phase-lock view, semantic time τ advances through collapse ticks, while imaginary time iT is unresolved phase rotation when collapse fails. The iT paper describes imaginary time as the record of unresolved, tension-loaded interpretive cycles that have not yet collapsed into discrete understanding; it also states that when collapse fails, the wavefunction keeps rotating in θ-space and no semantic tick occurs.

So:

```
(8.1) τ = admitted collapse / trace time.
(8.2) iT = unresolved pre-collapse phase memory.
```

The cold-atom paper experimentally realizes the τ side:

```
(8.3) entropy exchange → trace → τ.
```

But it also suggests where iT-like structure would live:

```
(8.4) unadmitted dark-sector relation → residual depth → possible future trace.
```

8.2 Two SMFT views of imaginary time

SMFT now has two complementary views of imaginary time.

View A — Phase-lock view

```
(8.5) iT = unresolved phase rotation before collapse.
```

This is the field-internal view. It looks at what happens when the wavefunction does not collapse into trace.

View B — Admissibility-depth view

```
(8.6) iTime_P = FilterDepth(PreTimeField_P).
```

This is the protocol / ledger view. It looks at what must be filtered before possibility becomes trace.

The Residual-to-Ledger article makes this explicit: pre-time is relation-rich possibility before ordered trace; iTime is admissibility depth by which pre-time possibility becomes filterable; real time is the order of admitted trace.

So the full chain is:

```
(8.7) PreTimeField_P → iTime_P → Trace_P → Ledger_P → Time_P.
```

or, in cold-atom language:

```
(8.8) Dark residual → barrier admissibility → entropy trace → bright ledger → entropic time.
```

8.3 Dark-sector residual as iT-like depth

The dark sector is not literally SMFT iT. It is a physical subsystem. But relative to the bright-sector ledger, it plays an iT-like role:

```
(8.9) It is structured.
(8.10) It is not fully visible.
(8.11) It can affect future bright-sector trace.
(8.12) It becomes time-relevant only when admitted through the gate.
```

Therefore:

```
(8.13) iT_like,bright = residual depth before bright-sector trace.
```

The important phrase is **relative to the bright sector**.

The same physical state can be:

```
(8.14) ledgered for one protocol,
(8.15) residual for another protocol,
(8.16) pre-time for a future declared world.
```

This is exactly why the Residual-to-Ledger framework says the pre-time field of one world may be the residual of another.

8.4 Entropic τ as released iT-like depth

When entropy crosses the barrier and enters the bright-sector record, something structurally similar to SMFT collapse occurs:

```
(8.17) unadmitted potential becomes admitted trace.
```

So the transition is:

```
(8.18) ResidualDepth → EntropyTrace → τ.
```

This lets us define:

```
(8.19) Entropic τ = ledgered release of residual depth.
```

Again, this is an interpretation, not the paper's own claim.

The paper's own experimentally grounded claim is that entropic time is operationally constructed from measurable entropy exchange and can order the bright-sector dynamics.

SMFT adds the ontology:

```
(8.20) Entropy exchange becomes time only when it becomes admitted trace.
```

---

9. What the Experiment Teaches SMFT

9.1 A laboratory analogue of declared time

The declared-disclosure framework says:

```
(9.1) Only after declaration can there be projection.
(9.2) Only after projection can there be gate.
(9.3) Only after gate can there be trace.
(9.4) Only after trace can there be ledger.
(9.5) Only after ledger can there be time.
```

The declaration article states this chain explicitly in its conclusion and defines time as the order of the declared disclosure operator applied to the pre-collapse field.

The cold-atom experiment provides a physical analogue:

```
(9.6) Declare bright sector.
(9.7) Measure bright-sector variables.
(9.8) Gate exchange through optical barrier.
(9.9) Record entropy-bearing events.
(9.10) Order records by τ.
(9.11) Recover internal dynamics.
```

This is why the experiment is so valuable for SMFT.

It shows that ledger-time is not merely a philosophical metaphor. In at least one controlled physical analogue, an internally useful time can be constructed from boundary-mediated trace.

9.2 SMFT gains an operational test language

The experiment suggests how SMFT-style time claims might be made more testable.

Instead of saying:

```
(9.12) Time is semantic collapse.
```

one can ask:

```
(9.13) What is the declared subsystem?
(9.14) What is the boundary?
(9.15) What is the gate?
(9.16) What counts as trace?
(9.17) What remains residual?
(9.18) Does the trace order reproduce observed dynamics?
```

This converts SMFT from an abstract ontology into an experimental diagnostic template.

For the cold-atom paper, the answers are unusually clear:

SMFT test question	Cold-atom answer
Declared subsystem?	bright sector
Boundary?	optical barrier
Gate strength?	barrier height V
Trace variable?	entropy-bearing bright-sector observables
Residual?	dark sector
Internal time?	τ from entropy exchange
Dynamics reproduced?	yes, via entropic-time Schrödinger equation

The paper concludes that cold atoms provide a controlled environment for benchmarking relational time constructions and entropy-based arrows of time.

SMFT can now say:

```
(9.19) This is also a controlled environment for benchmarking ledger-time constructions.
```

9.3 The cold-atom result disciplines SMFT

The experiment also disciplines SMFT by warning against vague claims.

Not every hidden sector creates time.

Not every boundary creates a world.

Not every residual becomes seed.

The Residual-to-Ledger article makes this caution explicit: a hidden interior does not become a world merely by being hidden; it requires a boundary, a compact law-like grammar, trace capacity, and a way for pre-time possibility to become ordered history.

The cold-atom experiment supports this discipline. The dark sector does not automatically generate bright-sector time. Only when coupling permits entropy exchange

does bright-sector entropic time flow.

So:

```
(9.20) Residual + gate + trace capacity → possible ledger-time.
```

But:

```
(9.21) Residual alone ≠ time.
```

This is essential for keeping the theory mature.

9.4 The strongest lesson

The strongest lesson is this:

```
(9.22) Internal time is not a property of an isolated object alone.
(9.23) Internal time is a property of a declared object-plus-boundary-plus-trace system.
```

The bright sector has time because:

```
(9.24) it is bounded;
(9.25) it exchanges;
(9.26) it records;
(9.27) it orders;
(9.28) it evolves under the order it records.
```

This is the cleanest SMFT reading of the experiment.

Continue from Section 10: Relational Time, Thermal Time, and SMFT Ledger-Time.

10. Relational Time, Thermal Time, and SMFT Ledger-Time

10.1 Three ways to make time internal

The cold-atom paper sits at the intersection of several approaches to time.

The first is **relational time**:

```
(10.1) One part of a system functions as a clock for another part.
```

The second is **thermal or entropic time**:

```
(10.2) Thermodynamic structure supplies an arrow or flow parameter.
```

The third is **SMFT ledger-time**:

```
(10.3) Ordered admitted trace supplies the internal time of a declared world.
```

The paper explicitly compares its construction with the **thermal time hypothesis**, while noting an important difference: its entropic time is constructed operationally from measurable entropy exchange between subsystems, rather than from the algebraic structure of observables alone.

SMFT can clarify the relationship:

Framework	What makes time?	Main strength
Relational time	correlation between internal variables	removes need for external clock
Thermal time	state / entropy structure	gives thermodynamic direction
Entropic time	measurable entropy exchange	experimentally operational
SMFT ledger-time	admitted trace under declaration	explains why a subsystem has history

The cold-atom paper shows that entropic time can work as a relational internal clock. SMFT then interprets that success as a special case of ledger-time.

10.2 Entropic time as relational time after trace admission

The paper's entropic time is not just a thermodynamic scalar. It is relational because it orders the bright sector by its exchange with the dark sector.

In SMFT notation:

```
(10.4) EntropicTime_bright = order(Trace_bright | DarkResidual, Gate_V).
```

This means:

```
(10.5) The bright sector does not own time alone.
(10.6) It generates internal time through its relation to the dark sector.
```

But this relation is not enough by itself. The relation must produce trace.

Therefore:

```
(10.7) Entropic time = relational time after trace admission.
```

This sentence captures the SMFT upgrade.

Relational time says:

```
(10.8) One variable orders another.
```

SMFT says:

```
(10.9) One declared ledger orders admitted trace from boundary-mediated relation.
```

That is more specific, and more operational.

10.3 Why thermal time is not yet ledger-time

The thermal time hypothesis associates time flow with thermodynamic or algebraic state structure. The cold-atom paper notes the conceptual similarity but

distinguishes its own construction because the entropic time is directly built from experimentally measured entropy exchange.

SMFT would say:

```
(10.10) Thermal time becomes ledger-time only when thermal structure becomes admitted trace for a declared subsystem.
```

This distinction matters.

A thermal state may contain an abstract flow. But a world-history requires:

```
(10.11) boundary;
(10.12) observables;
(10.13) gate;
(10.14) trace rule;
(10.15) residual rule;
(10.16) ordered ledger.
```

The cold-atom paper supplies all of these.

That is why it is so valuable for SMFT.

It does not merely say entropy can define time. It shows **where entropy is measured, how it crosses a boundary, what variables become trace, and how that trace orders dynamics**.

10.4 Ledger-time as the missing interface

The three concepts can now be stacked:

```
(10.17) Relational time = correlation order.
(10.18) Entropic time = entropy-flow order.
(10.19) Ledger-time = admitted-trace order.
```

In the cold-atom experiment:

```
(10.20) τ_entropic = entropy-flow order.
```

In the SMFT interpretation:

```
(10.21) τ_entropic = admitted entropy-trace order.
```

Thus, SMFT does not reject relational or entropic time. It gives them a more general interface:

```
(10.22) InternalTime_P = order(AdmittedTrace_P).
```

Entropy is one powerful way to create admitted trace. But in other systems, trace may be legal, cognitive, biological, institutional, computational, or cosmological.

This is how the cold-atom paper becomes more than a cold-atom result. It becomes a template for studying world-time.

---

11. The Experiment as a Physical Analogue of Declared Worldhood

11.1 A world is a protocol-bound subsystem

SMFT defines a world not as a metaphysical absolute, but as a protocol-bound region of disclosed structure.

A world requires:

```
(11.1) Boundary_P.
(11.2) Projection_P.
(11.3) Gate_P.
(11.4) Trace_P.
(11.5) Residual_P.
(11.6) Ledger_P.
```

Then:

```
(11.7) World_P = Boundary_P + Gate_P + Trace_P + Residual_P + Ledger_P.
```

and:

```
(11.8) Time_P = order(Ledger_P).
```

The cold-atom experiment gives a minimal physical example:

```
(11.9) World_bright = Barrier + BrightProjection + EntropyGate + EntropyTrace + DarkResidual + BrightLedger.
```

The bright sector is therefore world-like because it is not merely a spatial half of the condensate. It is a half of the condensate under a declared observation protocol.

11.2 The bright sector's time is not borrowed

One might say:

```
(11.10) The experiment still uses lab time, so the internal time is not real.
```

But that misses the point.

The lab uses external time to measure and validate the system. The paper's internal question is different:

```
(11.11) Can the bright sector's events be ordered using variables internal to the partitioned system?
```

The paper answers yes, within the constructed analogue: entropic time orders the bright-sector dynamics and supports an effective equation that reproduces observed evolution.

SMFT reads this as:

```
(11.12) External validation time ≠ internal ledger-time.
```

The bright sector's time is not "borrowed" from lab time once the internal ordering variable is constructed. Lab time remains the researcher's validation frame. Entropic τ becomes the bright-sector's internal ordering frame.

This matches SMFT's protocol-relative view:

```
(11.13) Time depends on the declared ledger.
```

Different protocols may produce different time variables. That is not a weakness. It is the point of relational time.

11.3 Declared worldhood without metaphysical inflation

The cold-atom bright sector is not conscious. It is not a semantic mind. It is not literally an observer in the human sense.

But it does instantiate a minimal declared world structure:

```
(11.14) boundary;
(11.15) internal variables;
(11.16) exchange with residual;
(11.17) traceable entropy;
(11.18) ordered internal dynamics.
```

SMFT does not need to anthropomorphize the system. The point is structural.

The experiment shows that a subsystem can acquire a self-contained ordering parameter when its boundary exchange is converted into trace.

That is enough to make it a physical analogue of ledgered worldhood.

---

12. Implications for the Problem of Time

12.1 The problem may be partly a problem of declaration

The Wheeler–DeWitt problem asks how time can emerge when the total description lacks an external temporal parameter.

SMFT suggests that the problem of time may partly be a problem of **undeclared totality**.

If we look only at the total closed field, we may have no internal clock. But once we declare a subsystem, boundary, observables, gate, trace rule, and residual rule, a time variable may emerge.

In compact form:

```
(12.1) Totality alone may be timeless.
(12.2) Declared subsystem + trace ledger may be time-bearing.
```

The cold-atom experiment realizes this pattern.

The total condensate is treated as effectively closed. The bright sector, however, becomes time-bearing through entropy exchange with the dark sector.

Thus:

```
(12.3) Time may not belong to the total field first.
(12.4) Time may belong to declared subsystems that can maintain ordered trace.
```

This is a profound shift.

12.2 Internal time as boundary-generated history

In the experiment, the bright sector's time is not derived from arbitrary labeling. It is derived from boundary-mediated entropy exchange.

SMFT generalizes this:

```
(12.5) InternalTime_P = BoundaryGeneratedHistory_P.
```

More explicitly:

```
(12.6) InternalTime_P = order({events admitted through Gate_P and recorded by TraceRule_P}).
```

This means internal time is not merely local motion. It is **history generated by a boundary**.

The boundary decides what can enter. The gate regulates the exchange. The trace rule records what matters. The ledger orders the record. The resulting order is time for that declared world.

12.3 The arrow of time as trace asymmetry

Fundamental physical equations are often time-symmetric. The arrow of time is usually associated with entropy increase, coarse-graining, and special boundary conditions.

The cold-atom paper enters this conversation by constructing an internal entropic arrow inside a globally well-isolated system. It verifies that the whole mini-universe entropy is constant within error bars, while the bright sector has entropy flow that can define internal time.

SMFT interprets this as:

```
(12.7) The arrow of time is not merely entropy increase.
(12.8) The arrow of time is trace asymmetry under a declared ledger.
```

Entropy flow matters because it creates a recordable asymmetry for the bright sector.

Thus:

```
(12.9) Arrow_P = asymmetry of admitted trace under protocol P.
```

In this view, entropy is not reduced to "disorder." Entropy becomes a measure of **trace-bearing exchange**.

12.4 Why global purity and local time can coexist

The paper begins from the tension that a total pure state or closed system may conserve fine-grained entropy, while a subsystem may show reduced or coarse-grained entropy flow.

SMFT has a natural way to express this:

```
(12.10) GlobalField may remain closed.
(12.11) LocalWorld_P may still generate ledger-time.
```

There is no contradiction because the two descriptions belong to different protocols.

For the total field:

```
(12.12) Residual and trace are not yet split.
```

For the bright-sector world:

```
(12.13) Trace and residual are split by the barrier and observation rule.
```

Time appears only after the split.

So:

```
(12.14) Time is not a property of totality alone.
(12.15) Time is a property of a declared trace/residual split.
```

This is one of the most elegant SMFT interpretations of the experiment.

---

13. Implications for SMFT, AI, Cognition, and Cosmology

13.1 Implication for SMFT

The experiment gives SMFT a concrete template:

```
(13.1) Define subsystem.
(13.2) Declare boundary.
(13.3) Identify residual.
(13.4) Define gate.
(13.5) Measure trace.
(13.6) Construct internal time.
(13.7) Test whether dynamics can be rewritten in that time.
```

This is a possible general research protocol for SMFT-inspired work.

Instead of making broad claims about time, one can ask:

```
(13.8) What is the trace variable?
(13.9) What is the residual?
(13.10) What gate converts residual into trace?
(13.11) Does the trace order reproduce dynamics?
```

The cold-atom paper answers these questions in a clean physical setting.

13.2 Implication for AI agents

AI systems usually process sequences by token order or wall-clock runtime. But an SMFT-inspired agent may need a stronger form of time: **ledger-time**.

For an AI agent:

```
(13.12) Prompt tokens are not yet agent time.
(13.13) Tool calls are not yet agent time.
(13.14) Memory entries are not yet agent time unless they become trace-bearing constraints.
```

An AI agent's internal time would be closer to:

```
(13.15) AgentTime = order(AdmittedMemoryTrace).
```

The cold-atom paper suggests a useful analogy:

Cold-atom system	AI agent
bright sector	active context
dark sector	unobserved memory / latent context / external world
barrier	retrieval gate / tool gate / memory gate
entropy exchange	information update
trace	committed memory / state update
entropic time	agent ledger-time

Thus, an AI agent becomes more world-like when it can decide what enters memory, what remains residual, and how admitted trace constrains future action.

This is exactly the SMFT view of observerhood.

13.3 Implication for cognition

Human subjective time is not identical to clock time.

Some periods pass without memorable trace. Other moments become dense, irreversible, and identity-shaping. This fits the SMFT distinction:

```
(13.16) External duration ≠ subjective ledger-time.
```

A person may experience:

```
(13.17) boredom = low trace density;
(13.18) trauma = excessive unresolved residual / blocked integration;
(13.19) flow = high coherence between action, trace, and internal rhythm;
(13.20) insight = sudden admission of residual into trace.
```

The cold-atom experiment is physical, not psychological. But structurally, it supports the general principle:

```
(13.21) Internal time depends on what the system can record as consequential change.
```

That principle may apply across physical, cognitive, institutional, and AI systems, provided each domain defines its boundary, gate, trace, and residual carefully.

13.4 Implication for cosmology

The cosmological extension must remain speculative.

Still, the experiment suggests an interesting grammar:

```
(13.22) Cosmic time may be ledger-time for a declared world-region.
```

If the universe as totality is timeless in a Wheeler–DeWitt-like description, then experienced cosmic time may arise from internal partitions, trace formation, decoherence, entropy flow, and observer-compatible records.

SMFT would phrase this as:

```
(13.23) CosmicTime_P = order(CosmologicalTrace_P).
```

The speculative Residual-to-Ledger extension would add:

```
(13.24) Pre-time is relation-rich residual before a world's trace ledger begins.
```

The cold-atom paper does not prove this. But it gives a disciplined analogue:

```
(13.25) A closed system can contain a subsystem whose time emerges from boundary-mediated entropy trace.
```

That is enough to make the cosmological question sharper.

---

14. Limits of the SMFT Interpretation

14.1 The experiment does not prove SMFT

This must be stated clearly.

The cold-atom paper demonstrates an operational entropic internal time in a controlled analogue system. It does not prove:

```
(14.1) semantic fields are physically fundamental;
(14.2) imaginary time is literally semantic phase memory;
(14.3) the universe is generated by residual-to-ledger cycles;
(14.4) black holes are semantic phase-lock zones.
```

Those are broader SMFT interpretations or speculative extensions.

The correct claim is:

```
(14.5) The experiment is structurally compatible with SMFT's ledger-time ontology.
```

14.2 The dark sector is protocol-relative residual

The dark sector is not residual in an absolute sense. It is residual relative to the bright-sector observation protocol.

Under a different protocol, the dark sector could itself be observed and ledgered.

Therefore:

```
(14.6) Residual_P is protocol-relative.
```

This is not a weakness. It is central to SMFT.

A thing is not residual because it is unreal. It is residual because it is outside the current ledger.

14.3 Entropic time is one kind of ledger-time, not all time

The article should not identify all time with entropy.

SMFT should instead say:

```
(14.7) Entropic time is one experimentally grounded form of ledger-time.
```

Other systems may use other trace variables:

Domain	Possible trace variable
cold atoms	entropy exchange
biology	metabolic work / structural maintenance
law	admissible evidence and judgment
AI	committed memory / verified state
cognition	remembered experience
institutions	official records
cosmology	decohered records / entropy-bearing events

The general form is:

```
(14.8) Time_P = order(Trace_P).
```

Entropy is one important trace type, not the only one.

14.4 Analogue is not identity

The paper uses a cold-atom analogue of minisuperspace-like relational time. The bright sector is not literally a universe; its big bang and big crunch are not literal cosmological singularities. The author explicitly presents the platform as an experimentally grounded testbed for relational-time constructions and related analogue questions, not as a direct reproduction of quantum cosmology.

SMFT should preserve that discipline.

Thus:

(14.9) Analogue similarity ≠ ontological identity.

But analogues matter because they test structural grammar.

The experiment tests whether:

(14.10) boundary + entropy exchange + internal variables can produce usable internal time.

That is exactly the structural grammar SMFT cares about.

---

15. Conclusion: Time Begins When a World Can Keep Trace

The cold-atom paper demonstrates that a well-isolated many-body system can be partitioned into an observed bright sector and an unobserved dark sector, and that the observed sector can be ordered by an internally constructed entropic time derived from measurable entropy exchange. The resulting internal time orders the bright-sector dynamics and supports an effective Schrödinger equation capable of reproducing the measured evolution.

SMFT interprets this as a physical analogue of ledger-time.

The core chain is:

(15.1) Total field → declared subsystem → boundary gate → residual exchange → trace → ledger → time.

For the cold-atom system:

(15.2) Total condensate → bright sector → optical barrier → dark-sector exchange → entropy trace → entropic ledger → bright-sector time.

The deepest lesson is not merely that entropy can function as a clock.

The deeper lesson is:

(15.3) A subsystem's time is not everything that happens around it.
(15.4) A subsystem's time is what it can admit, record, and order as its own history.

This is why the experiment matters for SMFT. It supplies a laboratory analogue of a central ontological thesis:

(15.5) Time is ordered admitted trace.

Entropy exchange becomes time only when it crosses the boundary, passes the gate, becomes measurable, enters the ledger, and constrains the order of the subsystem's world.

The bright sector begins its world not when the total condensate exists, but when trace enters. It closes its world not when reality ends, but when trace-bearing structure withdraws. Between episodes, lab time may pass, but no bright-sector entropic time need elapse.

In final form:

(15.6) NoTrace_P ⇒ NoTime_P.
(15.7) Trace_P ⇒ Ledger_P.
(15.8) Ledger_P ⇒ Time_P.

Or in ordinary language:

> **Time begins not when something changes, but when a world can admit change into trace and keep it as history.**

Appendix A — Cold-Atom to SMFT Translation Table

Cold-atom paper	SMFT / ledger interpretation	Comment
Total isolated BEC	Σ_total	closed field-like system
Bright sector	Σ_P, declared observed world	what the protocol chooses to observe
Dark sector	Residual_P	unledgered remainder relative to bright sector
Optical barrier	Boundary + Gate	separates and regulates admission
Barrier height V	gate strength / admissibility threshold	controls exchange rate
$\hat{H}$_coupling	cross-boundary coupling	residual-to-trace channel
Atom transfer	boundary-crossing event	candidate trace
Entropy exchange	entropy-bearing trace admission	basis of internal time
Absorption images	projection record	measurement trace
N	bright-sector population	amount of admitted material
ϕ	clock-like internal variable	non-global due to recollapse
Σ / a	scale degree of freedom	size of bright world-expression
S	entropy of bright sector	trace density / thermodynamic record
τ	entropic ledger-time	order of entropy-bearing traces
Big bang	first trace admission	start of ledger episode
Big crunch	ledger episode closure	return toward residual
Heat death	trace stagnation	no internal entropic time flow
Eq. (6)	dynamics in ledger-time	residual-modulated world law

Appendix B — Equation Bridge

B.1 Academic structure

(B.1) $\hat{H} = \hat{H}_{bright} + \hat{H}_{dark} + \hat{H}_{coupling}$.

(B.2) $\tau(\lambda) = \sigma/k_B \int_\lambda (dS/d\varphi)|d\varphi|$.

(B.3) $i\hbar\partial_\tau\psi(\tau,a) = \Phi(\tau)\psi(\tau,a) + \Lambda(\tau)H_{geom}\psi(\tau,a)$.

B.2 SMFT declaration structure

(B.4) P = (B, Δ, h, u).

(B.5) $\Sigma_P = Declare(\Sigma_0 \mid q, \phi, P)$.

(B.6) $\mathfrak{D}_P = \text{UpdateTrace}_P \circ \text{Gate}_P \circ \hat{O}_P \circ \text{Declare}_P$.

(B.7) $\text{Time}_P = \text{order}(\mathfrak{D}_P(\Sigma_0))$.

B.3 Cold-atom ledger form

(B.8) $\Sigma_{\text{total}} = \Sigma_{\text{bright}} \cup \Sigma_{\text{dark}} \cup \text{Gate}_V$.

(B.9) $\text{Trace}_{k,\text{bri}} = \text{Gate}_V(\text{EntropyExchange}_k)$.

(B.10) $L_{k+1,\text{bri}} = \text{Update}(L_{k,\text{bri}}, \text{Trace}_{k,\text{bri}})$.

(B.11) $\tau_{\text{bri}} = \text{order}(L_{\text{bri}})$.

B.4 Residual-to-ledger form

(B.12) DarkResidual → Gate_V → EntropyTrace → BrightLedger → BrightTime.

(B.13) NoTrace_bri ⇒ NoTime_bri.

(B.14) EntropicLedgerTime_bri = order(EntropyTrace_bri).

Continue with Appendix C — Clock Quality Criteria, Appendix D — Imaginary Time / Entropic Time / Ledger Time, and Appendix E — Experimental Predictions Inspired by SMFT.

Appendix C — Clock Quality Criteria

C.1 Why clock quality matters

The cold-atom paper shows that not every changing internal variable is a good clock.

The scalar-like variable ϕ changes during the bright-sector evolution, but in a recollapsing system it is not globally monotonic. It can reverse direction and therefore become ambiguous as a global ordering parameter. The paper therefore defines an entropy-weighted internal time τ that can order the bright-sector dynamics more robustly.

SMFT generalizes this into a clock-quality principle:

(C.1) A clock is not merely a changing variable.
(C.2) A clock is a trace-ordering variable.

A variable becomes time-like only when it can help a declared subsystem keep ordered history.

C.2 Four basic criteria

A candidate internal clock should satisfy at least four criteria:

(C.3) ClockQuality = Monotonicity + Traceability + BoundaryRelevance + Reproducibility.

Where:

Criterion	Meaning	Cold-atom example
Monotonicity	It must order events without frequent ambiguity.	τ grows almost everywhere; ϕ alone can reverse.
Traceability	It must correspond to recorded, measurable change.	τ is built from measured entropy S.
Boundary relevance	It must reflect the declared subsystem's relation to its environment/residual.	τ depends on entropy exchange between bright and dark sectors.
Reproducibility	It must work across repeated runs or controlled parameter changes.	τ is tested for multiple barrier heights V.

This allows a simple ranking:

(C.4) GoodClock_P = variable that orders Trace_P robustly under protocol P.

C.3 Why ϕ is not enough

The paper initially treats ϕ as an analogue scalar field. But because the bright sector expands and recollapses, ϕ is not globally monotonic.

In SMFT language:

(C.5) φ = state variable.
(C.6) $\varphi \neq$ guaranteed ledger variable.

A state variable can describe where the system is, but it may not preserve a direction of history.

Thus:

(C.7) Change_φ does not necessarily imply Time_P.

This is why entropy is needed. Entropy exchange supplies a trace-bearing asymmetry.

C.4 Why entropy works better here

Entropy works because it is tied to boundary-mediated exchange.

In the experiment:

(C.8) $S_{\text{bright}} = N \cdot s$.

where **N** is the number of atoms in the bright sector and **s** is entropy per atom. The paper notes that since entropy per particle remains of order unity, total entropy is effectively proportional to the number of atoms in the bright sector, linking entropy flow directly to atom-number dynamics.

SMFT reads this as:

(C.9) Entropy flow = measurable trace of boundary admission.

Therefore:

```
(C.10) τ_entropic is not just a mathematical reparameterization.
(C.11) τ_entropic is a trace-ordering variable tied to the bright sector's declared boundary.
```

C.5 Extended clock-quality formula

A stronger SMFT clock-quality expression is:

```
(C.12) ClockQuality_P(C) = M_P(C) + T_P(C) + B_P(C) + R_P(C) + I_P(C).
```

Where:

```
(C.13) M_P(C) = monotonicity under protocol P.
(C.14) T_P(C) = trace density captured by C.
(C.15) B_P(C) = boundary relevance.
(C.16) R_P(C) = reproducibility across runs.
(C.17) I_P(C) = invariance under admissible reframing.
```

This gives a general SMFT diagnostic:

```
(C.18) A clock is strong when it continues to order admitted trace under changes of protocol that should not change the underlying history.
```

For the cold-atom paper, future work could compare:

```
(C.19) ClockQuality(φ).
(C.20) ClockQuality(S).
(C.21) ClockQuality(N).
(C.22) ClockQuality(Σ).
(C.23) ClockQuality(τ_entropic).
```

The expectation is that **τ_entropic** should outperform **ϕ** as a global internal ordering variable in recollapsing regimes.

Appendix D — Imaginary Time, Entropic Time, and Ledger Time

D.1 Three different "times"

The broader SMFT framework distinguishes at least three time-like concepts:

Time concept	Basic meaning	Status
Imaginary time / iT	unresolved pre-collapse or pre-admission depth	pre-ledger
Entropic time τ_entropic	entropy-exchange ordering variable	experimental internal clock
Ledger time Time_P	order of admitted trace under protocol P	general SMFT time

They are related but not identical.

D.2 Imaginary time as phase-lock memory

The SMFT phase-lock article defines imaginary time as unresolved semantic phase rotation when collapse fails. In that view, real semantic time **τ** advances through collapse ticks, while **iT** accumulates when the wavefunction continues rotating in θ-space without being collapsed by an observer projection.

In compact form:

```
(D.1) Collapse succeeds → τ tick.
(D.2) Collapse fails → iT accumulates.
```

Or:

```
(D.3) iT = unresolved phase memory before trace.
```

This is the **field-internal** view of imaginary time.

D.3 Imaginary time as admissibility depth

The later admissibility-depth view reformulates imaginary time in ledger terms:

```
(D.4) iTime_P = admissibility depth before trace.
```

This means iTime is not a hidden clock ticking before time. It is the depth, cost, filtering, or unresolved relation that must be processed before something can become ledgerable.

In the residual-to-ledger vocabulary:

```
(D.5) PreTimeField_P → iTime_P → Trace_P → Ledger_P → Time_P.
```

So:

```
(D.6) iTime belongs to the unadmitted side.
(D.7) Time belongs to the admitted side.
```

D.4 Entropic time in the cold-atom paper

The cold-atom paper constructs entropic time as:

```
(D.8) τ(λ) = σ/k_B ∫_λ (dS/dφ)|dφ|.
```

This **τ** is not SMFT iT. It is closer to **ledger time**, because it orders the bright-sector events after entropy exchange becomes measurable.

Therefore:

```
(D.9) τ_entropic = order of admitted entropy trace.
```

The experiment directly constructs the admitted side:

```
(D.10) entropy exchange → trace → τ_entropic.
```

It does not directly observe the pre-admitted side:

```
(D.11) hidden residual depth → iT-like structure.
```

D.5 Where iT-like depth lives in the experiment

Relative to the bright-sector protocol, iT-like depth would correspond to structure that is:

```
(D.12) outside the bright-sector ledger;
(D.13) not yet admitted through the barrier;
(D.14) still capable of affecting future bright-sector trace;
(D.15) hidden in the dark sector or in unmeasured correlations.
```

Thus:

```
(D.16) iT_like,bright = ResidualDepth_bright before entropy trace admission.
```

This is not a claim that the dark sector literally **is** imaginary time. It is a structural interpretation:

```
(D.17) The dark sector plays the role of pre-ledger residual relative to the bright-sector world.
```

D.6 The conversion chain

The full SMFT interpretation of the cold-atom paper is:

```
(D.18) DarkResidual
→ Gate_V
→ EntropyAdmission
→ Trace_bright
→ Ledger_bright
→ τ_entropic.
```

Adding imaginary time:

```
(D.19) iT_like depth
→ admissibility crossing
→ trace
→ τ.
```

In words:

> **Imaginary time is the unadmitted depth of possible trace. Entropic time is that depth after it crosses a boundary and becomes ordered record.**

D.7 Three-time comparison table

Question	iT / admissibility depth	Entropic time	Ledger time
What is it?	unresolved pre-trace depth	entropy-based internal order	order of admitted trace
Does it require trace?	no	yes	yes
Is it experimentally measured in the cold-atom paper?	not directly	yes	structurally yes
What is its SMFT role?	residual tension / pre-time	physical trace-clock	general world-time
Cold-atom analogue	dark-sector residual / hidden correlations	τ from entropy exchange	bright-sector history

---

Appendix E — Experimental Predictions Inspired by SMFT

E.1 Why predictions matter

The SMFT interpretation becomes more valuable if it suggests future experimental questions.

The cold-atom paper already proposes future directions, including comparing multiple internal clocks, investigating singularities and bouncing scenarios, testing reversibility with Loschmidt echo, engineering analogue black holes, and realizing tunnelling scenarios between sectors.

SMFT adds a specific diagnostic angle:

```
(E.1) Test how boundary, gate, trace, residual, and ledger jointly determine internal time.
```

E.2 Prediction 1 — Barrier height should act as ledger-rate control

The paper already observes that increasing barrier height reduces entropy exchange and slows entropic time.

SMFT predicts this more generally:

```
(E.2) ∂τ/∂t_lab should decrease as Gate_V becomes more restrictive.
```

In ledger language:

```
(E.3) Gate restriction lowers trace-admission rate.
(E.4) Lower trace-admission rate slows ledger-time.
```

A future test could fit:

```
(E.5) LedgerRate_bright ≈ f(GateTransparency_V, EntropyGradient, CouplingStrength).
```

The exact function is experimental, but the qualitative direction is clear:

```
(E.6) More admissible exchange → faster internal ledger-time.
(E.7) Less admissible exchange → slower internal ledger-time.
```

E.3 Prediction 2 — Different trace definitions should produce different internal times

The paper defines entropy using a particular experimental method. SMFT predicts that different trace rules may produce different internal times.

For example:

```
(E.8) τ_S = time from entropy trace.
(E.9) τ_N = time from particle-number trace.
(E.10) τ_Σ = time from scale-width trace.
(E.11) τ_corr = time from correlation trace.
```

These may not be equivalent.

SMFT would ask:

```
(E.12) Which trace rule produces the most stable ledger?
```

That can be tested by comparing which internal time best reproduces the observed dynamics under changes in barrier height, interaction strength, and initial state.

E.4 Prediction 3 — Clock disagreement reveals residual structure

If two internal clocks disagree, this is not merely error. It may reveal residual.

For example:

```
(E.13) Δτ_{S,N} = τ_S − τ_N.
```

SMFT interpretation:

```
(E.14) ClockDisagreement = ResidualSignature.
```

If entropy time and particle-number time diverge, the divergence may indicate hidden correlations, coarse-graining loss, or unmeasured structure in the dark sector.

Thus:

```
(E.15) residual is not just noise;
(E.16) residual is what prevents all clocks from agreeing.
```

This is a very useful experimental diagnostic.

E.5 Prediction 4 — Loschmidt echo should separate reversible dynamics from irreversible ledgering

The paper proposes future reversibility tests through Loschmidt echo.

SMFT predicts a distinction:

```
(E.17) Dynamical reversal ≠ ledger reversal.
```

A Loschmidt echo may reverse certain physical degrees of freedom, but the trace ledger may not be fully erased if measurement, coarse-graining, or residual coupling has already produced irreversible record.

Possible outcomes:

Result	SMFT interpretation
Dynamics reverse and τ reverses	weak ledger / nearly reversible trace
Dynamics reverse but τ does not	ledger irreversibility
τ partially rewinds	partial trace reversibility
residual grows after attempted reversal	hidden trace leakage

Key test:

```
(E.18) Does reversing dynamics reverse admitted trace, or only state motion?
```

This would directly probe the SMFT distinction between state evolution and ledger history.

E.6 Prediction 5 — Measurement feedback should thicken ledger-time

If measurement is not passive but changes the system, then stronger measurement feedback should increase trace density.

SMFT predicts:

```
(E.19) Stronger projection Ô_P → stronger trace formation → altered internal time.
```

In cold atoms, this could correspond to different measurement protocols, imaging frequencies, or feedback controls.

Possible SMFT prediction:

```
(E.20) Increasing measurement-induced trace should modify τ_entropic if the measurement back-reacts on entropy exchange.
```

This would connect the cold-atom experiment more directly to observer-induced ledger formation.

E.7 Prediction 6 — A second bright sector should create competing ledgers

One could partition the condensate into more than one observed sector:

```
(E.21) Σ_total = Σ_bright1 ∪ Σ_bright2 ∪ Σ_dark ∪ Gates.
```

Then each bright sector may have its own ledger-time:

```
(E.22) τ_1 = order(Trace_1).
(E.23) τ_2 = order(Trace_2).
```

SMFT predicts that the relative synchronization of these internal times should depend on shared residual and gate coupling.

Possible tests:

```
(E.24) τ_1 ≈ τ_2 when sectors share similar exchange.
(E.25) τ_1 diverges from τ_2 when gate structure differs.
```

```
(E.26) synchronization events occur when residual exchange becomes common trace.
```

This could become a laboratory analogue of cross-observer time agreement.

E.8 Prediction 7 — Artificial black-hole analogues should act as trace traps

The paper suggests engineering analogue black holes in the bright sector using shaped attractive potentials.

SMFT predicts that such regions may act as:

```
(E.27) trace traps;
(E.28) residual accumulators;
(E.29) local ledger slow zones.
```

If the analogue black-hole region traps degrees of freedom in a way that reduces trace admission to the rest of the bright sector, then internal time outside and inside the trap may desynchronize.

A possible diagnostic:

```
(E.30) τ_inside / τ_outside should depend on trace escape rate.
```

This could create an experimentally tractable analogue of horizon-related ledger separation.

Appendix F — Short Glossary

Term	Meaning in this article
Admitted trace	A change that passes the gate and becomes recorded under a protocol.
Bright sector	Observed subsystem in the cold-atom paper; SMFT declared world.
Clock quality	How well a variable orders trace as internal time.
Dark sector	Unobserved subsystem; residual relative to bright-sector ledger.
Declaration	Protocol that specifies boundary, observables, gate, trace, and residual.
Entropic time	Internal time built from entropy exchange in the cold-atom paper.
Entropic ledger time	SMFT interpretation of entropic time as ordered entropy trace.
Gate	Boundary rule controlling what becomes admissible trace.
Imaginary time / iT	Pre-ledger unresolved depth; in SMFT phase view, unresolved phase rotation before collapse.
Ledger	Ordered trace structure that gives a world internal history.
Ledger-time	Time defined as the order of admitted trace.
Residual	Unadmitted, unresolved, or unledgered remainder under a protocol.
Trace	Recorded event that constrains future interpretation or dynamics.
World_P	A declared, protocol-bound world with boundary, gate, trace, residual, and ledger.

Final Closing

The cold-atom paper is valuable because it makes a difficult philosophical problem experimentally concrete. It does not merely discuss whether time can be relational. It builds a system where internal time can be constructed, measured, tested, and used to reproduce dynamics.

SMFT adds a compact interpretation:

```
(F.1) The bright sector has time because it has a ledger.
(F.2) It has a ledger because entropy exchange becomes trace.
(F.3) Entropy exchange becomes trace because the barrier acts as a gate.
(F.4) The dark sector matters because residual feeds the gate.
```

Therefore:

```
(F.5) Internal time = ordered admitted trace.
```

The experiment shows this in a physical analogue:

```
(F.6) No entropy exchange → no internal entropic time.
(F.7) Boundary-mediated entropy exchange → internal entropic time.
(F.8) Ordered trace → effective dynamics.
```

This is why the paper fits SMFT so well.

It gives experimental shape to one of SMFT's deepest claims:

> **A world does not have time because an outside clock says so.**
> **A world has time when it can keep trace of what its boundary admits into history.**

© 2026 Danny Yeung. All rights reserved. 版权所有 不得转载

Disclaimer

This book is the product of a collaboration between the author and OpenAI's GPT 5.5, Google AI, Gemini 3, NoteBookLM, X's Grok, Claude' Sonnet 4.6 language model. While every effort has been made to ensure accuracy, clarity, and insight, the content is generated with the assistance of artificial intelligence and may contain factual, interpretive, or mathematical errors. Readers are encouraged to approach the ideas with critical thinking and to consult primary scientific literature where appropriate.

This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.